YASHPAL YADAV

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SUMMARY

B.Tech and MS by Research student at IIIT Hyderabad specializing in NLP, LLM reasoning, and table understanding. Experienced in full-stack and mobile development with published apps on App Store and Play Store. Proven research background in improving LLM numerical reasoning accuracy by 32% through tool-augmented systems.

EDUCATION

International Institute of Information Technology, Hyderabad **2022-2026**
B. Tech. + MS (Research) in CSE *Hyderabad, Telangana*

Banaras Hindu University **2018-2021**
B. Sc. (Hons.) in Physics *Varanasi, Uttar Pradesh*

TECHNICAL SKILLS

Languages: C/C++, Python, HTML/CSS, JavaScript, TypeScript, Dart, Kotlin, Swift, SQL
ML/AI: PyTorch, HuggingFace Transformers, LLMs (GPT-3.5, FlanT5), Prompt Engineering, scikit-learn
Data & Tools: Pandas, NumPy, Jupyter, Git, Postman, Android Studio
Frameworks & Infra: Flutter, React.js, React Native, Node.js, Express.js, MongoDB, Firebase, Redis, WebSockets, Linux

EXPERIENCE

Language Technologies Research Center **July 2024 - Present**
Undergraduate Researcher *IIIT Hyderabad*

- Developed and implemented a hybrid methodology to improve LLM (GPT-3.5, FlanT5) performance on numerical reasoning and structured table-based QA tasks.
- Engineered a tool-augmented system that allows an LLM to delegate arithmetic operations to an external Python "calculator" module, improving calculation accuracy from 60% to 92% on manipulation tasks.
- Designed and validated robust prompt-engineering strategies, demonstrating that a Pandas DataFrame-style (DFLoader) table representation achieved 88% accuracy on structure-understanding tasks, significantly outperforming CSV (71%) and HTML (60%) formats.
- Fine-tuned and evaluated models on the FeTaQA benchmark for free-form generative table QA, achieving a 28.5 BLEU score and reducing factual hallucinations to 5% by implementing a post-generation verification strategy.
- Created a comprehensive evaluation suite using "numerical probes" based on a 4-level reasoning taxonomy and tested zero-shot generalization on unseen medical data, achieving 70% accuracy.

SKDIV®Industries Private Limited **May 2024 - Nov 2025**
Frontend Developer Intern *Remote*

- App Store URL: `https://apps.apple.com/au/app/crew/id6739217561`
- Architected and implemented the "CREW Calling" feature, leveraging Flutter and the Livekit SDK to deliver high-quality, real-time audio communication for both 1-on-1 and group calls.
- Engineered the core real-time messaging frontend for the CREW platform using Flutter, implementing secure WebSocket connections to facilitate seamless exchange of text, images, and voice notes.
- Implemented robust offline capabilities by integrating IndexedDB with a Redis cache backend, ensuring users have uninterrupted access to recent conversations and significantly boosting application resilience.
- Enhanced user engagement by integrating Firebase Cloud Messaging (FCM) to deliver timely and reliable push and in-app notifications, ensuring critical updates are never missed.

Software Engineering Research Center **May 2023 - July 2024**
Undergraduate Researcher *IIIT Hyderabad*

- **Demo URL:** https://vlab-os.github.io/OS4all/
- Developed an interactive web-based platform, funded by the Ministry of Human Resource Development (MHRD), Government of India, for students across India to learn Operating System concepts.
- Led the development of *'Process Life Cycle'*, *'Context Switching'*, and *'Scheduling'* experiments using HTML, CSS, and JavaScript, resulting in a 20% increase in student engagement and understanding of complex OS concepts.
- Integrated Express.js for backend operations of the *'Process Life Cycle'* experiment, enhancing data analysis capabilities and reducing data processing time by 15%, leading to more efficient project outcomes.

HydrowVerse Pvt Ltd. **Jan 2023 - May 2023**
Project Intern *Hyderabad, Telangana*

- **Application URL:** https://play.google.com/store/apps/details?id=com.mycompany.hydrow
- Engineered the Hydrowverse app using *Flutter*, seamlessly integrating IoT devices via API calls to fetch real-time data from ThingSpeak, enhancing user experience and data accuracy by 30%.
- Designed intuitive interface using *Flutter* to provide users with live water level updates, daily usage tracking, and detection of potential wastage using "Starr" smart water level device.
- Established seamless device-phone connectivity via QR code scanning, enabling users to conveniently get notification alerts & access water resource summaries.

PROJECTS

Greddit — Reddit-like Social Platform | *MongoDB, Express.js, React.js, Node.js*

- GitHub: Yashpal-0/Greddit
- Engineered a full-stack Reddit-like web application using the MERN stack with real-time chat powered by MongoDB Atlas.

ML Algorithms from Scratch | *scikit-learn, Pandas, NumPy, Python, hmmlearn*

- Implemented 6 Machine Learning algorithms from scratch: PCA, K-means, GMM, Multinomial Logistic Regression/Classification, Multi Layer Perceptron, and Decision Trees.

CShell — Custom Linux Terminal | *C, Linux, System Calls*

- GitHub: Yashpal-0/CShell
- Developed a Linux Terminal in C, leveraging kernel system calls exclusively.
- Implemented command execution, process management (foreground/background), I/O redirection, and process termination.

xv6 Process Scheduler | *C, Kernel Programming, Linux, System Calls*

- GitHub: Yashpal-0/xv6-scheduling
- Developed system call tracing (`strace`), FCFS, PBS, and MLFQ schedulers by modifying core xv6 kernel modules.
- Enhanced process management with creation time, dynamic priorities, and priority queues.

ACHIEVEMENTS & LEADERSHIP

- Achieved a top 2% rank in JEE Mains 2018, out of 1.2 million candidates.
- Letter of Appreciation from Smriti Zubin Irani, MHRD minister, Govt. of India for excellent academic performance in High School.